

Support Vector Machines: A Lecture from First Principles

MFML Sec9

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August 23, 2025

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1 The Goal: Designing the Perfect Street

1.1 Introduction & The Core Question

Our journey begins with a simple goal: to separate two classes of data. Imagine you are a city planner tasked with building a road to separate two neighborhoods, “Red Town” and “Blue Town”. If the towns are **linearly separable**, you can draw a straight line between them.

However, there are infinitely many possible lines you could draw. Which one is the *best* one?

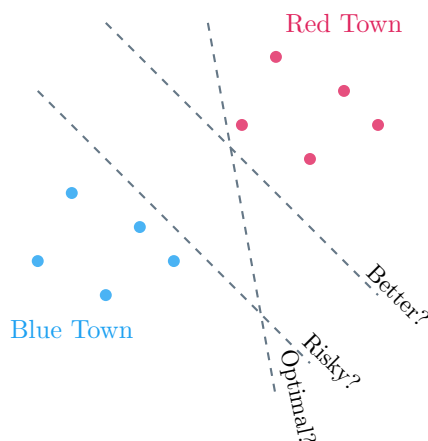


figure Multiple possible lines can separate the data, but which is optimal?

Analogy

The Widest Street A road that skims too close to one neighborhood is risky. A small error in a house’s location, or a new house being built, could put it on the wrong side. The best road is the one that is maximally far from the closest houses in *both* neighborhoods. This creates the largest possible “safety buffer”. In machine learning, this buffer is the **margin**, and our entire goal is to find the classifier that creates the **maximum margin**.

1.2 Formalizing the Blueprint: The Primal Problem

Let’s translate this “widest street” analogy into a precise mathematical blueprint.

1.2.1 Step 1A: Defining the Street’s Direction ($\mathbf{w}$)

We define the center line of our street with the equation $\mathbf{w}^T \mathbf{x} + b = 0$. The vector $\mathbf{w}$ is a **normal vector**, meaning it is perpendicular (at a 90° angle) to the line.

Quick Proof: Take any two points, $\mathbf{x}_A$ and $\mathbf{x}_B$, on the center line. They both satisfy the equation, so $\mathbf{w}^T \mathbf{x}_A + b = 0$ and $\mathbf{w}^T \mathbf{x}_B + b = 0$. Subtracting one equation from the other gives $\mathbf{w}^T (\mathbf{x}_A - \mathbf{x}_B) = 0$. The vector $(\mathbf{x}_A - \mathbf{x}_B)$ lies *along* the line. Since its dot product with $\mathbf{w}$ is zero, $\mathbf{w}$ must be orthogonal to the line.

1.2.2 Step 1B: Setting a Standard for Our Sidewalks (+1 and -1)

Next, we define the edges of our street—the sidewalks where the closest houses will lie. For mathematical elegance, SVM makes a clever choice called the **canonical hyperplane**:

- **Positive Sidewalk:** $\mathbf{w}^T \mathbf{x} + b = 1$
- **Negative Sidewalk:** $\mathbf{w}^T \mathbf{x} + b = -1$

This is a **normalization**. We can scale $\mathbf{w}$ and b by any factor without changing the center line, so we have the freedom to make this simplifying choice.

1.2.3 Step 1C: Deriving the Street's Width (The Margin Formula)

Now we can calculate the street's width—the margin—from first principles. The margin is the perpendicular distance between the two sidewalks.

Let $\mathbf{x}_1$ be a point on the positive sidewalk ($\mathbf{w}^T \mathbf{x}_1 + b = 1$) and $\mathbf{x}_2$ be a point on the negative sidewalk ($\mathbf{w}^T \mathbf{x}_2 + b = -1$). The vector connecting these points is $(\mathbf{x}_1 - \mathbf{x}_2)$. The margin is the length of the projection of this vector onto the normal vector $\mathbf{w}$.

$$\begin{aligned}\text{Margin} &= \frac{|\mathbf{w}^T (\mathbf{x}_1 - \mathbf{x}_2)|}{\|\mathbf{w}\|} \\ &= \frac{|\mathbf{w}^T \mathbf{x}_1 - \mathbf{w}^T \mathbf{x}_2|}{\|\mathbf{w}\|}\end{aligned}$$

Since $\mathbf{w}^T \mathbf{x}_1 = 1 - b$ and $\mathbf{w}^T \mathbf{x}_2 = -1 - b$:

$$\begin{aligned}&= \frac{|(1 - b) - (-1 - b)|}{\|\mathbf{w}\|} \\ &= \frac{|1 - b + 1 + b|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|}\end{aligned}$$

Key Result

This is our crucial result: **The margin width is cleanly defined as $\frac{2}{\|\mathbf{w}\|}$.**

1.2.4 The Formal Primal Problem

We can now formally state our goal.

- **Objective:** To maximize the margin $\frac{2}{\|\mathbf{w}\|}$, we must **minimize** $\|\mathbf{w}\|$. For mathematical convenience (to make the calculus cleaner), we minimize $\frac{1}{2}\|\mathbf{w}\|^2$ instead.
- **Constraints:** We must ensure every house ($\mathbf{x}_i$) with label $y_i \in \{+1, -1\}$ is on the correct side of the street and *off the pavement*. The single, elegant rule for this is:

$$y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

Definition

The SVM Primal Problem This gives us our formal **Primal Problem**:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad \forall i \quad (1)$$

2 The Optimization Toolkit

2.1 The Lagrangian: A Tool for Solving with Rules

Our primal problem has constraints, which makes it difficult to solve directly. We use the **Method of Lagrange Multipliers** to incorporate the constraints into the objective function.

Analogy

The Site Inspector Imagine you, the city planner, hire a “site inspector” (we’ll call them $\alpha_i \geq 0$) for each house. The inspector’s job is to levy a fine if their assigned house is on the pavement or on the wrong side of the street. Your new goal is to build the widest street possible *while also minimizing the total fines you have to pay*.

We first write our constraint in the standard form $g_i(\mathbf{w}, b) \leq 0$:

$$1 - y_i(\mathbf{w}^T \mathbf{x}_i + b) \leq 0$$

The **Lagrangian** function L is then:

$$L(\mathbf{w}, b, \alpha) = \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{\text{Original Objective}} - \sum_i \underbrace{\alpha_i}_{\text{Fine}} \underbrace{[y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1]}_{\text{Constraint Satisfaction}} \quad (2)$$

where α_i are the Lagrange multipliers.

2.2 The Magic of Duality

The Lagrangian creates a “minimax” game. We control $\mathbf{w}, b$ to minimize L , while an adversary controls α to maximize L . This gives us two perspectives:

- **The Primal Problem (P):** $\min_{\mathbf{w}, b} \max_{\alpha \geq 0} L(\mathbf{w}, b, \alpha)$
- **The Dual Problem (D):** $\max_{\alpha \geq 0} \min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha)$

For convex problems like SVM, **strong duality** holds, meaning the solutions are identical ($P^* = D^*$). We can solve the easier dual problem and get the correct answer.

3 The Rules of the Game: Slater, KKT, and Complementary Slackness

3.1 Slater’s Condition: The “Green Light” for Duality

Think of Slater’s condition as a quality check on an optimization problem. Before you go through the trouble of solving an easier “dual” problem, you need to be sure its answer will be the same as the original “primal” problem’s answer. Slater’s condition gives you that guarantee.

What it is: It’s a set of requirements for an optimization problem. For our purposes, the main conditions are that the problem must be **convex** (shaped like a bowl with a single minimum) and, crucially, that there must be at least one possible solution that is strictly inside the constraint boundaries (not right on the edge). This ensures the feasible region has some “interior volume”.

Why it matters: If a convex problem satisfies Slater’s condition, then **strong duality holds**. This is the “green light.” It guarantees that the optimal value of the primal problem is exactly equal to the optimal value of the dual problem. For SVMs, this is fantastic because the dual problem is often easier to solve and reveals the kernel trick.

For SVMs: The hard-margin SVM problem is a convex problem that satisfies Slater’s condition. This is why we can confidently switch to solving the dual problem, knowing the answer will be correct.

3.2 The KKT Conditions: The “Optimality Checklist”

If strong duality holds, the final, perfect solution to our problem must obey a specific set of rules. These are the Karush-Kuhn-Tucker (KKT) conditions. Think of them as an “optimality checklist” that the answer must satisfy. The four main rules are:

1. **Primal Feasibility:** The solution must obey the original rules of the problem. In SVM, this means all data points must be correctly classified and on or outside the margin.
2. **Dual Feasibility:** The penalties (Lagrange multipliers) must be non-negative. You can’t have a “negative” penalty for breaking a rule.
3. **Stationarity:** At the optimal solution, all the forces are perfectly balanced. Mathematically, this means the gradient of the Lagrangian is zero.
4. **Complementary Slackness:** This is the most insightful rule and gets its own section below.

3.3 Complementary Slackness: The “Magic Switch”

This is the most brilliant part of the KKT conditions. It’s the rule that connects each constraint directly to its penalty (its Lagrange multiplier, α_i in SVM).

The Core Rule: The rule states that for every data point i , the following product must be zero:

$$\alpha_i^* [y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1] = 0$$

The Two Scenarios: For this equation to be true, one of two things must happen for each data point:

- The point is **NOT on the margin**. The point is safely classified and far away from the “sidewalk”. This means the constraint is inactive, and the term in the brackets $[\cdot]$ is a positive number (not zero). For the whole product to be zero, its penalty α_i^* **must be 0**. The penalty is “switched off” because this point isn’t helping to define the boundary.
- The penalty α_i^* **is positive**. If the point has a positive penalty, it is considered important. For the whole product to be zero, the term in the brackets $[\cdot]$ **must be 0**. This means $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 = 0$, which tells us the point lies *exactly on the margin*.

Why it’s Crucial for SVM: This condition is the mathematical definition of a **Support Vector**. It mathematically proves that the only points that get a positive α_i value are the ones precisely on the margin. All other points get $\alpha_i = 0$. Since the final formula for the hyperplane ($\mathbf{w} = \sum \alpha_i y_i \mathbf{x}_i$) only depends on the points with non-zero α_i , this is how the SVM elegantly ignores the vast majority of the data and focuses only on the critical points—the support vectors—that define the solution.

4 The Construction: Solving the Dual Problem

To find the dual, we must first solve the inner part of the dual problem: $\min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha)$. We use the stationarity condition from KKT.

4.0.1 1. Derivative with respect to $\mathbf{w}$

$$\frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_i \alpha_i y_i \mathbf{x}_i = 0 \implies \boxed{\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i}$$

Insight: The optimal weight vector $\mathbf{w}$ is a linear combination of the training data points!

4.0.2 2. Derivative with respect to b

$$\frac{\partial L}{\partial b} = - \sum_i \alpha_i y_i = 0 \implies \boxed{\sum_i \alpha_i y_i = 0}$$

Insight: The weighted influence of positive and negative points must balance.

4.0.3 3. The Great Substitution

We substitute these two results back into the Lagrangian to eliminate $\mathbf{w}$ and b .

$$L = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_i \alpha_i y_i \mathbf{w}^T \mathbf{x}_i - b \sum_i \alpha_i y_i + \sum_i \alpha_i$$

The term with b is zero from our second result.

$$L = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \mathbf{w}^T \left(\sum_i \alpha_i y_i \mathbf{x}_i \right) + \sum_i \alpha_i$$

The term in parentheses is just $\mathbf{w}$ from our first result.

$$L = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \mathbf{w}^T \mathbf{w} + \sum_i \alpha_i = \sum_i \alpha_i - \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

Finally, substitute the definition of $\mathbf{w}$ entirely in terms of α .

$$\begin{aligned} L &= \sum_i \alpha_i - \frac{1}{2} \left(\sum_i \alpha_i y_i \mathbf{x}_i \right)^T \left(\sum_j \alpha_j y_j \mathbf{x}_j \right) \\ L &= \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^T \mathbf{x}_j) \end{aligned}$$

Definition

The SVM Dual Problem This gives us the final **SVM Dual Problem**:

$$\max_{\alpha} \left(\sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^T \mathbf{x}_j) \right) \quad (3)$$

subject to $\alpha_i \geq 0$ and $\sum_i \alpha_i y_i = 0$.

5 A Concrete Example: KKT in Action

Let's solve the problem: **Minimize** $f(x, y) = x^2 + y^2$ **subject to** $g(x, y) = x + y - 3 \leq 0$. The goal is to find the point closest to the origin within the allowed region.

5.1 1. Check Slater's Condition

First, we check if this problem has the “green light” for using duality.

- **Is the objective function convex?** Yes, $f(x, y) = x^2 + y^2$ is convex. It describes a paraboloid, which is shaped like a perfect bowl and has a single minimum.

- **Is the constraint function convex?** Yes, $g(x, y) = x + y - 3$ is a linear function, and all linear functions are convex.
- **Is there a point strictly inside the boundary?** The boundary is the line $x + y - 3 = 0$. We need to find any point (x, y) where $x + y - 3 < 0$. Let's test the origin, $(0, 0)$: $0 + 0 - 3 = -3$. Since $-3 < 0$, the point $(0, 0)$ is strictly inside the feasible region.

Conclusion: The problem satisfies all the requirements. Slater's condition holds, which guarantees that strong duality is in effect. The solution we find using the KKT conditions will be the true global minimum.

5.2 2. Write the KKT Checklist

The optimal solution (x^*, y^*) and its penalty λ^* must satisfy:

1. **Primal Feasibility:** $x + y - 3 \leq 0$.
2. **Dual Feasibility:** $\lambda \geq 0$.
3. **Stationarity:** $\nabla f + \lambda \nabla g = 0 \implies [2x, 2y] + \lambda[1, 1] = [0, 0]$. This gives us $2x + \lambda = 0$ and $2y + \lambda = 0$, which means $x = y = -\lambda/2$.
4. **Complementary Slackness:** $\lambda(x + y - 3) = 0$.

5.3 3. Solve using Complementary Slackness

This “magic switch” gives us two cases to check.

5.3.1 Case 1: The penalty is “off” ($\lambda = 0$)

This implies the solution is not on the boundary line.

- From stationarity: $2x + 0 = 0 \implies x = 0$. And $2y + 0 = 0 \implies y = 0$.
- Our candidate point is $(0, 0)$.
- Check feasibility: Does it satisfy $x + y - 3 \leq 0$? Yes, $0 + 0 - 3 = -3 \leq 0$.
- **Conclusion:** $(x = 0, y = 0, \lambda = 0)$ is a valid solution satisfying all KKT conditions. The objective value is $f(0, 0) = 0$.

5.3.2 Case 2: The solution is “on the boundary” ($x + y - 3 = 0$)

This allows the penalty λ to be active ($\lambda > 0$).

- From stationarity, we know $x = y$.
- Substitute this into the boundary equation: $x + (x) - 3 = 0 \implies 2x = 3 \implies x = 1.5$.
- Our candidate point is $(1.5, 1.5)$.
- Find its λ using stationarity: $2(1.5) + \lambda = 0 \implies 3 + \lambda = 0 \implies \lambda = -3$.
- **Conclusion:** This candidate is **invalid** because it violates dual feasibility (λ must be ≥ 0).

Key Result

The only point that satisfies the entire KKT checklist is $(\mathbf{x}, \mathbf{y}) = (\mathbf{0}, \mathbf{0})$. This makes perfect sense; the point in the allowed region that is closest to the origin is the origin itself. The KKT conditions provide a rigorous way to prove this.

6 Conclusion and a Look Ahead

Today, we built the Support Vector Machine model from a single intuitive idea. We started with the “widest street,” formalized it into the Primal Problem, used the powerful tools of the Lagrangian and Duality to find an easier path, and in solving it, discovered the beautiful and efficient concept of Support Vectors.

- **Soft Margins:** In the real world, data is rarely perfectly separable. We can introduce “slack variables” to allow for some misclassifications, leading to the Soft Margin SVM.
- **The Kernel Trick:** Notice the dual problem only uses the dot product of data points, $\mathbf{x}_i^T \mathbf{x}_j$. This allows us to replace it with a more complex “kernel” function, mapping the data to a higher dimension and creating powerful non-linear classifiers.